

A low-power daisy-chain controller implementation in BMS based on power mode switching

1st Xinhao Xu
*Electronic and information
engineering college
Tongji University
Shanghai, China
2130805@tongji.edu.cn*

2nd Yongzhen Chen*
*Electronic and information
engineering college
Tongji University
Shanghai, China
yzchen@tongji.edu.cn

3rd Jiangfeng Wu
*Electronic and information
engineering college
Tongji University
Shanghai, China
jfwu@tongji.edu.cn*

Abstract—This paper proposes a low-power daisy-chain interface controller that can be applied to a battery management system. Based on the power mode switching, the paper divides the chip into three power modes, SHUTDOWN, SLEEP and ACTIVE, and combine multi-voltage domains, clock gating, power gating and asynchronous circuits, RTL coding techniques to achieve dynamic switching of power consumption. The lowest supply currents can reach 3.81uA under 180nm process.

Index Terms—low-power, BMS, daisy-chain, multi-voltage, clock gating

I. INTRODUCTION

A typical battery pack for an electric vehicle typically consists of up to 96 lithium-ion cells connected in series, and the total voltage can exceed 400V. Therefore multiple analog front-end (AFE) sampling chips cascaded together in a daisy-chain structure are required to measure information such as battery voltage, current and temperature, and transmit the measurement results to the microcontroller (MCU) to implement a battery management system. However, there is no internationally developed unified solution for daisy-chain communication interface, and different communication solutions are given in the AFE chips launched by some international chip manufacturers. Some researchers have also used the standard SPI interface to form a daisy-chain[1, 2], but no low-power solution is given and the chip will continue to draw high currents.

Since AFE chips are usually powered directly by the battery pack they are measuring, a low-power design is important for AFE chips. Lower power consumption means better battery pack endurance. The paper will refer to TI's daisy-chain communication protocol[3] and propose a low-power solution for daisy-chain architectures, dividing the chip's power modes into SHUTDOWN, SLEEP and ACTIVE to avoid the power overhead associated with all circuits being in working operation all the time. The switching of the power modes can be flexibly controlled by the software. This paper is organized as follows. Section II describes the communication architecture of the daisy-chain chips as well as the basic communication protocols. Section III specifically gives the low-power design implemented in this paper. The results and conclusions are given in Sections IV and V, respectively.

II. ARCHITECTURE DESIGN AND PROTOCOL DEFINITION

Fig. 1 gives a schematic diagram of AFE chips connected in a daisy-chain ring structure. Each chip has three communication ports: UART, COMH and COML. UART is a full-duplex serial bus used to communicate with the MCU, while COMH and COML are inter-chip communication interfaces consisting of a pair of bidirectional differential lines. Due to the different voltage domains, isolated communication devices need to be added between the chips.

The host can access and control all devices via the UART interface of the connected chip. In the TI's protocol, the chips in the daisy-chain can be divided into three categories in terms of location: base device, stack device and top stack device, where the base device communicates with the MCU through the UART interface and communicates with the stack device through one of the daisy-chain ports. The stack device uses both daisy-chain ports to communicate with the upstream and downstream chips. The top stack device communicates with the downstream chip via one of the daisy-chain ports. The communication interface structure in the chip is given in Fig. 2, where the transceiver physical layer analog circuits for each of the COMH and COML ports are necessary, as well as the digital control circuits designed in this paper.

The types of signal on each communication ports can be divided into two categories. One is a specific signal used for power mode switching and the other is a data signal, consisting of byte frames. The power mode switching signals can be divided into four categories: wakeup, slp2act, shutdown, hwrst. The MCU can flexibly control the power mode of all daisy-chain chips through these four types of switching signals. The type of signal received by UART interface is called Ping which is a long low level signal that distinguishes the above four types of signals according to the duration of low level.

The power mode switching signal sent and received through the daisy-chain port is called Tone, which is a low frequency periodic pulse signal. Firstly it can be defined that when com*p is high and com*n is low, and after maintaining a period of time, it flips to the opposite level, which is a positive pulse and vice versa, as shown in Fig. 3(a). Thus a valid Tone signal can be distinguished by the polarity and number of pulses,

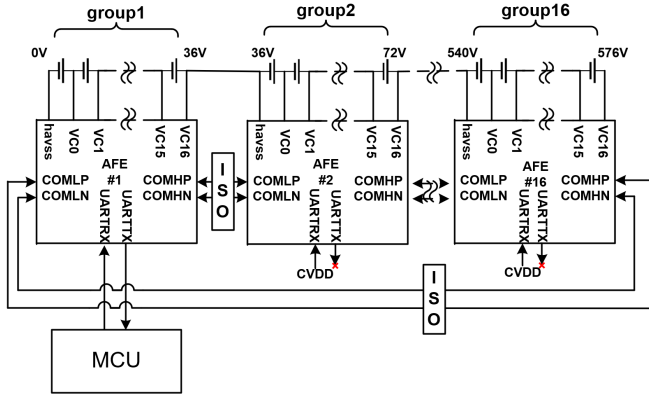


Fig. 1. AFE chips in a daisy-chain connected structure

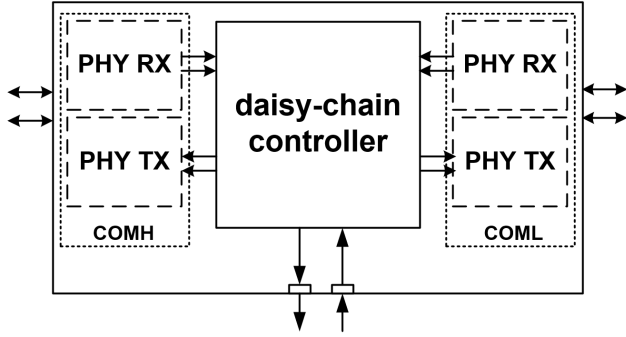


Fig. 2. Communication interface circuit structure

where the width of each pulse and the interval between the sent pulses is also limited, as shown in Fig. 3(b).

III. IMPLEMENTATION

In digital circuit design, circuit power consumption can be divided into dynamic power consumption and static power consumption. Dynamic power consumption is divided into internal power consumption and switching power consumption, related to the clock flip. Static power consumption is mainly caused by the leakage current power consumption. The power consumption expression is shown in Equation(1).

$$P = \frac{1}{2} \cdot C \cdot V_{DD}^2 \cdot f \cdot N_{SW} + Q_{SC} \cdot V_{DD} \cdot f \cdot N_{SW} + I_{leak} \cdot V_{DD} \quad (1)$$

We can find that the power consumption of the circuit is mainly related to the supply voltage V_{DD} and the circuit flip-flop frequency f . Therefore, from the system level, a multi-voltage domain design with two voltage domains, 5V and 1.8V is proposed. The 5V domain is used as an always-on power domain, where only a very small portion of the wake-up detection and control circuits are placed to realize the switching of the power modes. The 1.8V domain is used as a controlled voltage domain, where most of the functional

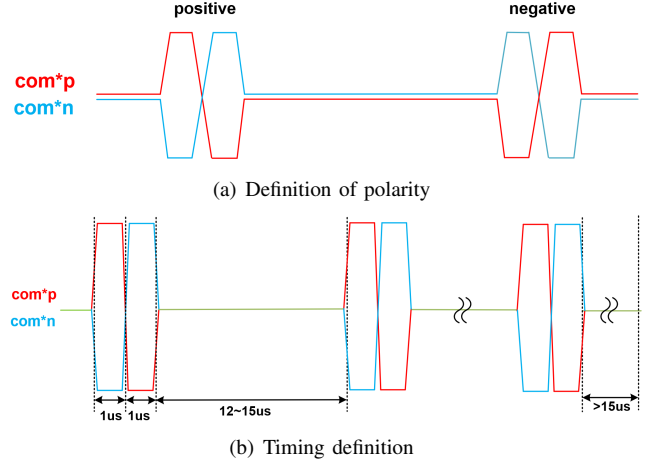


Fig. 3. Tone signal

circuits are placed, and is powered down in SHUTDOWN mode to minimize power consumption. Level shifters with isolation function need to be placed between the two voltage domains. Secondly, in order to reduce the circuit flip-flop frequency, there is a 32M high frequency oscillator and a 250K low frequency oscillator inside the chip, so that the circuits that do not have high performance requirements will work at low-frequency clock domain, in order to save dynamic power consumption. Furthermore, the 32M oscillator is disabled in SHUTDOWN mode and the 250K oscillator is always on. The system implementation block diagram is shown in Fig. 4.

Meanwhile, the state machine diagram of the power mode is shown in Fig. 5. After the power-on reset(POR) release, the chip enters the SHUTDOWN state, when only the 5V always-on power domain has power supply and the 1.8V power supply is turned off. If a wake-up signal is detected in the SHUTDOWN state, the chip enters the PREACTIVE state, in which the LDO for the 1.8V power is activated. After the 1.8V power supply has been stabilized, the chip enters the DEVRESET state, in which the circuits in the 1.8V voltage domain will be reset. After that, it enters the ACTIVE state, in which all the functions of the chip are enabled, and the chip can enter other power states according to the received Ping/Tone signals, or through the software configuration and the timeout mechanism. In the SLEEP state, the 1.8V power supply is retained, but the main function modules will be disabled by clock gatings to save the dynamic power. In the SLEEP state, there is also support for jumping to other power states based on Ping/Tone signals and the timeout mechanism. In the HWRST state, the POR circuit will be triggered to do a warm reset. The state machine circuit is implemented in the power_ctrl block in the 5V always-on power domain.

A. 5V always-on power domain circuit

In the 5v voltage domain, only a low frequency clock of 250K is used in order to minimize the dynamic power consumption in SHUTDOWN mode. But it is necessary to have the ability to detect the wake-up signal in SHUTDOWN

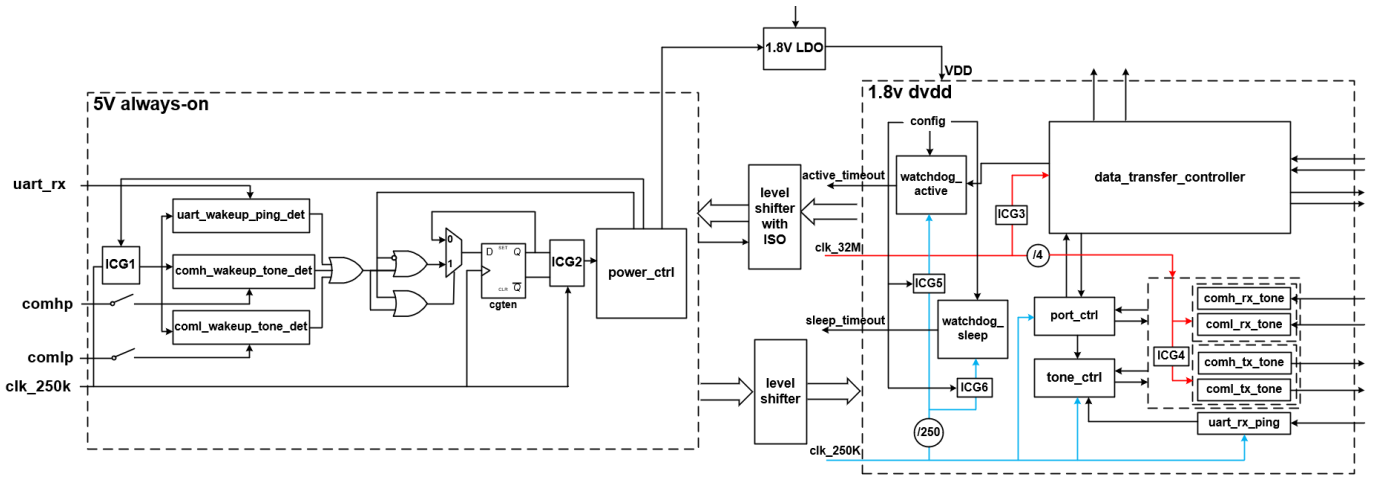


Fig. 4. daisy-chain controller implementation block diagram

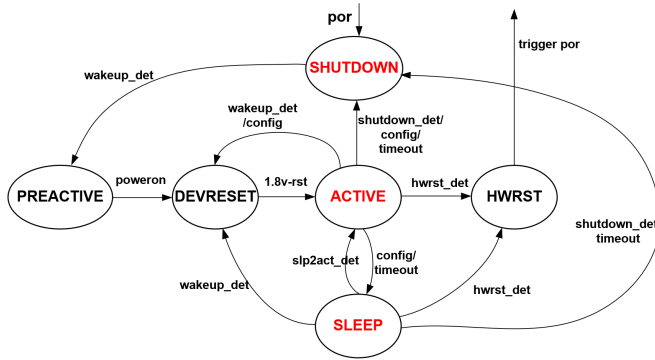


Fig. 5. Power mode state machine

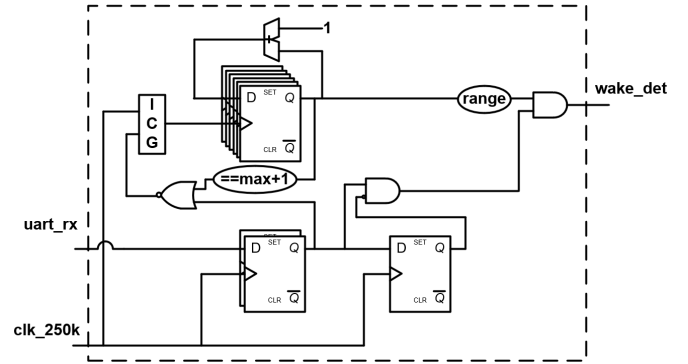


Fig. 7. Detection circuit in the uart_wakeup_ping_det block

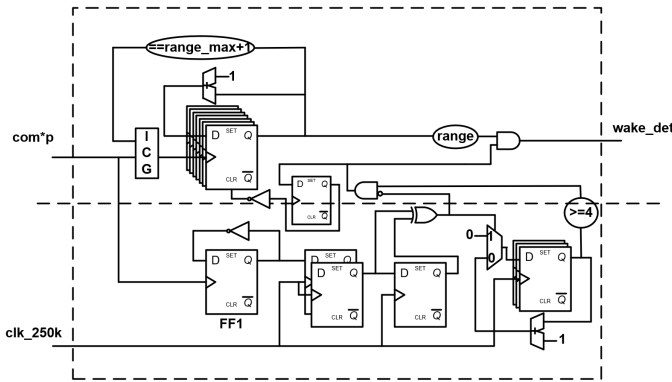


Fig. 6. Detection circuit in the comh/l_wakeup_tone_det block

mode, and a 250K clock cannot satisfy the detection of the pulse width requirement of the Tone signal in Fig. 3(b). So this work adopts an asynchronous circuit approach, connecting the com*p signal to the clock side of a set of registers, only detecting the flip-flop of the com*p signal and recording the number of pulses, the circuit structure is shown in Fig. 6.

The upper half of the circuit is a counter with com*p as the clock. When the counter value is equal to the maximum range plus one, it can be decided that it is not a valid Tone signal, then ICG is turned off and counting is stopped. The lower half of the circuit implements the detection of the Tone signal pulse interval. The FF1 register uses com*p as the clock to make the Q side flip, and the rest of the circuit uses 250K as the main clock. When the FF1 register does not flip for more than the maximum time, the Tone signal is considered to be received. Then we determine whether the number of received pulses is within the range, if it is, then output the wake_det signal and clear the com*p counter at the same time. Besides, as can be seen in Fig. 4, since the com*p signal acts as a clock drive registers, there are switches to be added at the input level to avoid the power overhead associated with the asynchronous registers in non-SHUTDOWN mode.

The way of detecting the wake-up signal from uart_rx is simply using the 250K clock to count when uart_rx is low and determining if the counter value is within the time range when uart_rx goes high. Clock gating is also inserted for the clock of counters consisting of a number of registers to turn off

the clock when uart_rx goes high or the count value exceeds the maximum range. The circuit diagram is shown in Fig. 7.

From a big view, as shown in Fig. 4, when the chip is in the SHUTDOWN mode, ICG1 is turned on and the three modules detecting wake-up signals will work. When a wakeup signal is detected, the Q side of the cgten register will update to 1 in the next cycle, which turns on the clock of the power_ctrl block. The shutdown state is then jumped out and ICG1 will be turned off. The clocks of the three modules that detect wake-up signals are turned off. When the chip returns to the SHUTDOWN mode, a pulse will be generated from the power_ctrl block and the Q side of the cgten register will be updated to 0. The clock of the power_ctrl block will be turned off, and ICG1 will be turned back on. The modules that detect the wake-up signals return to work. This design utilizes two clock gating in the block level to minimize the dynamic power consumption in the 5V power domain in both the SHUTDOWN and non-SHUTDOWN modes.

B. 1.8V dvdd power domain circuit

In the 1.8V power domain, as shown in Fig. 4, most of the functional circuits is implemented with two clock inputs, 32M and 250K. The data_transfer_controller block is responsible for receiving and sending data frames according to the protocol, and the whole circuit works in the 32M clock domain in order to achieve better performance. The blocks that receive and send the Tone signal work in the 8M clock domain which is obtained by quadrature of the 32M clock. The uart_rx_ping block detects the long low level Ping signal on the UART interface. The tone_ctrl block implements the control of Tone signal forwarding, and the port_ctrl block controls the port enable based on the chip's position in the daisy-chain and current power mode. All these three blocks work in the 250K clock domain. In addition, the 250K clock is divided by 250 to get a very low frequency clock of 1KHz to the watchdog_active and watchdog_sleep block, which implement timeout detection in ACTIVE and SLEEP modes according to the configuration.

In the ACTIVE mode, ICG3 and ICG4 are turned on, ICG6 is turned off. ICG5 is turned on depending on whether the timeout detection under ACTIVE mode is enabled by the current configuration. Therefore, the power consumption in the ACTIVE state is large, especially when there are a large number of communication packets for transmission. In the SLEEP mode, ICG3, ICG4 and ICG5 are turned off and the function of data transfer and sending Tone signal will be disabled. If the function of timeout detection under SLEEP mode is enabled, ICG6 will be turned on, otherwise it will be turned off.

Many clock gating techniques are used in the 1.8V power domain, and the clock frequency is reduced to the minimum considering the actual requirement, which largely reduces the dynamic power consumption overhead. At the same time, the RTL code gives priority to the use of coding techniques that can automatically generate clock gating to further reduce the power consumption caused by clock flip-flop.

IV. RESULTS

We have synthesized this design in 180nm process and obtained the final gate-level netlist file after the placement and routing design. Table I gives the power consumption of this design in three power modes by using the netlist.

TABLE I
POWER CONSUMPTION OF DAISY-CHAIN CONTROLLER

Power domain	Power consumption	power state		
		SHUTDOWN	SLEEP	ACTIVE
5V	Static power(W)	5.42e-07	5.38e-07	5.38e-07
	Dynamic power(W)	1.85e-05	3.21e-05	3.21e-05
1.8V	Static power(W)	0	1.79e-06	1.74e-06
	Dynamic power(W)	0	1.46e-04	6.95e-03
Total	Supply Currents(mA)	3.81e-03	8.86e-02	3.86

There are trade-offs in our design. In order to reduce the dynamic power consumption, we use low frequency clock in some control blocks to reduce the performance of the system. Meanwhile, we have to increase the area to add clock gatings and implement the clock domain crossing circuit between the low frequency clock and the high frequency clock. However, these sacrifices to reduce the power consumption of the system are totally acceptable in the BMS application scenario.

V. CONCLUSION

This paper successfully uses the power mode switching method to dynamically reduce the power consumption of the system, allowing the MCU to dynamically and flexibly control the power consumption of the chip through the daisy-chain interface. Under 180nm process, the daisy-chain interface controller designed in this paper has the smallest total supply currents of 3.81uA in SHUTDOWN mode, the largest of 3.86mA in ACTIVE mode, and 88.6uA in SLEEP mode.

REFERENCES

- [1] Xiaofei Wang, Hong Zhang, Long Zhang, Jianrong Zhang, and Yue Hao. A daisy-chain spi interface in a battery voltage monitoring ic for electric vehicles. In *2014 12th IEEE International Conference on Solid-State and Integrated Circuit Technology (ICSICT)*, pages 1–3, 2014.
- [2] Qidong Zhang, Yintang Yang, Changchun Chai, Institute Of Microelectronics, and Xidian University. A high ems daisy-chain spi interface for battery monitor system. *Journal of Semiconductors*, 2017.
- [3] Texas Instruments. Functional safety-compliant automotive 16s/14s/12s battery monitor, balancer and integrated hardware protector. Accessed on: October 21, 2022. [Online]. Available: https://www.ti.com/lit/ds/symlink/bq79616-q1.pdf?ts=1684153938731&ref_url=https%253A%252F%252Fwww.ti.com%252Fproduct%252FBQ79616-Q1.